

Intelligent mobile networks

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Major advances have been made in mobile communications in recent years. The first generation analogue cellular systems (TACS, NMT) are now giving way to the more advanced second generation digital systems (GSM). Both first and second generation systems have provided intelligent capabilities within the network, the second generation systems in particular offering far more advanced service capabilities than contemporary fixed networks. Developments also in fixed communications have seen intelligence added via the intelligent network (IN). The next (third) generation of mobile systems will provide convergence towards a common architecture, with more advanced capabilities for both fixed and mobile being provided by integrated intelligent network architectures. The paper introduces mobile communications and describes cellular systems aspects including the GSM network architecture. The requirements and possible architectures for the forthcoming third generation mobile system are described, illustrating the developments needed for intelligent networks.

1. Introduction to mobile systems

The past few decades have seen a massive increase in the use of mobile communications. Cellular telephones are commonplace within society and can be seen in use in all areas of modern life — at home, work and in public. The true dawn of mass market mobile communications is only around the corner. By the year 2000 it is expected that the small hand-held pocket communicator will be in use by a significant proportion of the population.

Mobile communications has been made possible by advances that have enabled compact radio technology and computing intelligence to be added to user's terminals and telecommunications networks. Telecommunications switches have evolved to include computer databases and complex interfaces to handle interactions with radio-equipped base-stations. Functionality has been added to enable the networks to track, locate and provide continuous communications for the users while they are mobile. Added intelligence in the terminal has enabled the traditional wire-line link to the switch to be replaced via radio. Intelligent schemes for optimising the use of the precious radio resources has also enabled high user-traffic densities to be achieved.

2. Analogue systems and the cordless telephone

The origin of land mobile communications appeared around 30—40 years ago with vehicle-mounted radio transmitters. Mobile communications have made massive leaps forward since then, with developments being split between two basic areas.

- 'cellular', where a telecommunications network is used to enable continuous communications over a wide area,
- 'cordless', where a more concentrated local coverage is provided over small 'pockets' — cordless is generally provided independent of the fixed network [1].

2.1 Analogue cellular

The contemporarily established first generation mobile systems within the majority of Europe are based upon analogue cellular technology, which is now starting to roll over towards digital technology. The common systems include the North American based AMPS (advanced mobile phone service), the Scandinavian NMT (Nordic mobile telephone) and TACS (total access communications system) within the UK, which is a variant of the American AMPS. Analogue speech transmission with frequency modulation operating in the 450 or 900 MHz bands is used for these analogue systems with the network components based around proprietary manufacturer's or national standards. With no internationally agreed network or radio interface standard, network vendors and operators had little flexibility in the production and procurement of network components. This problem has been resolved by the development of second generation digital systems. Within Europe the second generation standard for cellular communications is called GSM — the global system for mobile communications [2].

2.2 Cordless communications

Within the cordless area, developments were driven following the development of simple analogue cordless telephones. The early 80s saw the potential flooding of cheap imported units to a wide variety of frequency standards which offered tetherless communications from the handset to the base unit. Moves were made within Europe to develop standardised cordless communications. The initial development using digital communications was CT2 (Cordless Telephone 2), and offered improved quality and capacity on the radio path with a common air interface enabling handsets to 'roam' between cordless systems. The DECT (digital European cordless telecommunications) system included the support of ISDN and LAN data capabilities in excess of 384 kbit/s full duplex. CT2 operates around 900 MHz while the DECT standard was developed to operate at around 1.8 GHz. CT2 and DECT standards do not define the network aspects for the system; however, DECT does include the capability to support wide area mobility, if linked by a mobility supporting network such as GSM.

3. Cellular system aspects

The term 'cellular' is derived from the realisation of the system. Cellular comprises a number of transceivers geographically dispersed to give separate coverage areas or 'cells'. The type of layout produced optimises the radio spectrum resources by enabling radio frequencies to be re-used in non-adjacent cells. Terminal mobility is the feature of cellular systems that is the major difference from fixed telephony. Cellular systems offer mobile users the capability to make and receive calls from their mobile telephone while within coverage of the network. The cellular system automatically tracks and records the location of the terminal within the system to enable mobile terminating calls to be delivered. Once calls are in progress between the terminal and the network, communications can continue while the mobile terminal is moving geographically. This can mean the terminal crossing radio coverage boundaries (moving between cells) and requires a 'handover' to take place.

Mobility places major requirements upon the mobile network. The feature enabling the system to track the location of the subscribers is termed 'mobility management'. The link between the terminal and the network is subject to constant changes due to the requirements of the radio link. The quality of this radio link can change due to interference, path changes and system operations. The flexible management of the radio aspects to cater for this is termed 'radio resource management'.

To enable calls to be routed to the subscriber, the system must be able to locate the subscriber's terminal. The terminal will automatically inform the network of its current location via the 'location update' procedure as it moves

around within the radio coverage offered by the system. The network groups cells into location areas, the cells within the location areas broadcast location area identity information which the terminal then returns to the network, along with its own identity. From this information the network database can update the database entry of the subscriber with the current location. For a mobile terminating call, the dialled digits of the called subscriber will be used by the network to query the network database. The network will obtain the current location information held within the database and use this to initiate the paging procedure. The network will broadcast the subscriber's identity throughout the current location area. For a successful paging attempt the terminal will respond, and communications between the terminal and the network can be initiated. The network then uses this point of communications to deliver the call via internal routing.

In the case of a mobile originated call, the subscriber's terminal will attempt to make a call. The network will check the subscriber's database information to ensure the call may be made. Then, subject to any restrictions, the call will be allowed to continue as normal.

Due to the mobile nature of the subscriber, terminals may move during communication, possibly requiring the serving cell to change. To cater for this the handover procedure (also termed 'hand-off') is used to enable the call to pass between cells without 'dropping out'. The handover may require a change of both radio path and network path which should not noticeably interrupt communications.

Cellular systems place major requirements upon the networks which require added intelligence and processing capabilities to manage handover, mobility management and the subscriber's call-handling profile details.

4. The global system for mobile communications

As introduced earlier, nearly all of the countries within Europe currently offer some form of mobile telecommunications. The demand for a European-wide standard mobile communications system was created by the limitations of operators' early analogue systems. National solutions had led to expensive non-standard network and terminal equipment. The diversity of these national standards and the limitations on users being able to roam within mainland Europe (to make and receive calls in different countries), in conjunction with varying quality, coverage and capabilities, encouraged the start of developments of this system in 1982.

The result was GSM, the 'global system for mobile communications', which was initially developed within the 900 MHz band; however, the availability within the UK of spectrum at 1800 MHz enabled the scope of GSM to be extended to cover this band. The developments mainly

revised the radio aspects, with the system being referred to as DCS 1800 (digital cellular system 1800).

4.1 GSM services and features

The primary role of GSM was to provide mobile voice telephony. However, the system offers the capability of data services to complement ISDN at rates of up to 9600 bit/s full duplex. Other capabilities include the short message service (text messages), facsimile transmission and packet data access. The roaming capability throughout Europe is automatic — upon switching on the GSM phone in any area covered by GSM, the ‘visited’ network will notify the ‘home’ network of the location and obtain the subscriber’s parameters. Thus mobile originating and mobile terminating calls can be made.

Many of the early analogue systems encountered security problems with cloning of terminals and terminal theft. The GSM system has made significant improvements in the security area. The GSM subscription details are contained on a smartcard called the SIM (subscriber identity module), this has to be plugged or installed into the terminal before use. The GSM system provides digital encryption and authentication of communications, and can also check for type approval of terminals used on the network. Handset size developments have been a major factor in the success of GSM. Approximately 8 to 10 years were needed for analogue handsets to evolve from large ‘transportable terminals’ to pocketable phones. GSM terminals are already available in pocketable form despite being relatively new within the market-place. Figure 1 shows the progress made with GSM terminals. A GSM ‘SIM’ card is also shown.



Fig 1 GSM SIM card and terminals.

4.2 The GSM standard

As a standardised system GSM not only defined a common radio interface but also a network architecture which enabled network vendors to develop equipment to conform to the common standard and also enabled network operators to purchase equipment for use within the network from a variety of vendors. These network components should all interwork correctly due to conformance with the standard.

4.3 The GSM network architecture

The GSM network has been one of the major successes of European standardisation. Its standardised architecture has enabled a variety of options for manufacturers and operators to introduce GSM. The GSM architecture is shown in Fig 2. In many respects, the GSM architecture is a good example of an early intelligent network (IN) architecture as it follows the concept of separating switching and service-related functions.

Procedures for call handling, mobility management, handover, roaming and location management have all been defined using the architecture. The functionality of the various network entities is described below.

The **mobile station** (MS) consists of the physical equipment used by the GSM subscriber. The MS contains the subscriber identity module (SIM), which stores information such as the subscriber’s addressing information, authentication parameters and security functionality

The **air-interface** (Um) is the interface between the MS and the BSS.

The **base station system** (BSS) is the physical equipment providing the radio coverage. Functionally, the BSS is divided into two parts:

- the **base station controller** (BSC) which provides the control function and interfaces with the mobile switching centre (MSC) via the A interface, and is responsible for identifying the need for handovers and managing intra-BSC handovers,
- the **base transceiver station** (BTS) which provides the radio transmission functions for the radio coverage in the cell.

The **A-bis interface** between the two entities has been defined, enabling the two functions to be implemented separately, if required.

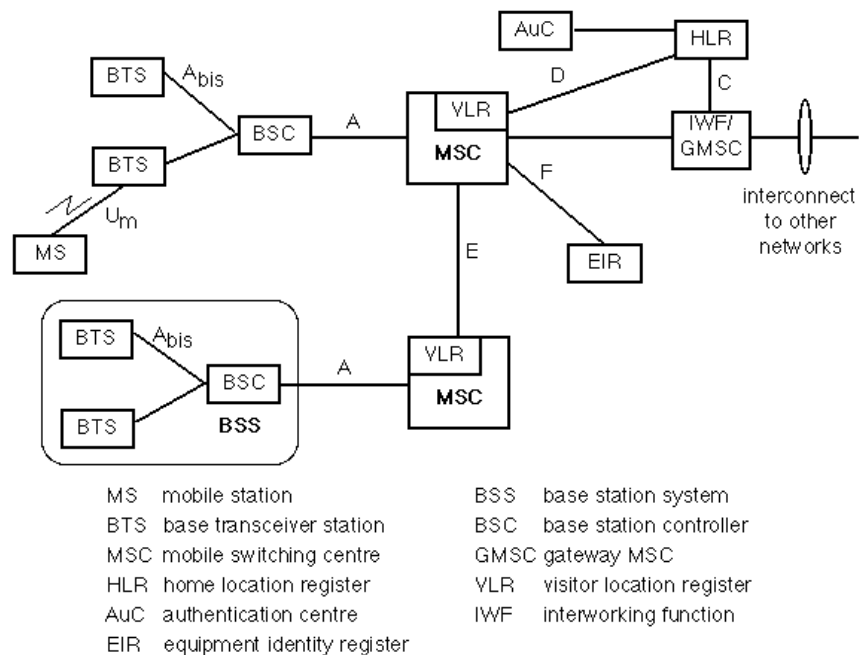


Fig 2 The GSM architecture.

The **A interface** is defined between the BSS and the MSC and carries information for the base station system application part (BSSAP) protocols. The BSSAP consists of two separate applications:

- the base station system management application part (BSSMAP) protocol which is used to support processes in the MSC and BSC,
- direct transfer application part (DTAP) which is used to relay messages between the MS and the MSC transparently.

The **mobile switching centre** (MSC) carries out the switching functions needed for all mobile stations within a geographical area controlled by the MSC. The MSC takes into account the mobile nature of the subscribers by managing the radio resources during the inter-BSC or inter-MSC handover process. It is also involved in the location update procedure, when the MS moves into a new area.

MSCs can have an interworking function (IWF) associated with them for interconnection purposes. The functionality of the IWF depends on the type of network to which the interconnection is required and also the services requested. Specially designated gateway MSCs (GMSCs) have interrogating node functionality which is used to access the HLR for routing and service related information for mobile terminating calls.

The C, D, E, and F network interfaces within the GSM network, shown in Fig 2, are based on the CCITT Signalling System Number 7 (SS No.7) and use the mobile application part (MAP) signalling procedures.

The interfaces between MSCs and the interconnection between the GSM network and other networks use standard connection-oriented protocols such as ISUP, TUP/NUP for call-handling purposes.

The **visitor location register** (VLR) is the database containing information needed to handle the calls set up or received by the MSs registered with the VLR. Within early versions of the GSM standard the 'B' interface was used between the MSC and the VLR, this interface has now been subsumed, with the VLR being integrated into the visited MSC. A mobile station roaming in an MSC area has its current location data managed by the visitor location register. The VLR is in charge of the data records of all the MSs registered within its catchment area. Certain requests involving the supplementary services that require additional information are met by the interrogation of the home location register (HLR).

The **home location register** (HLR) is the database responsible for the management of the mobile subscriber's records. It manages information related to service subscription profiles, e.g. teleservices, bearer services and the supplementary services that are to be made available to

the MS. It also dynamically stores the current location information such as the visited MSC/VLR of roaming mobile stations, allowing the appropriate routing of mobile terminating calls.

The **authentication centre** (AuC) is a database concerned with the security aspects. It is responsible for managing the allocation of individual MS's authentication parameters from the home network to the visited network. These are used by the visited network for authentication of the visiting MS and establishing ciphering on the radio channel.

The **equipment identity register** (EIR) is a database which stores equipment identities which are used by the GSM system to identify unique pieces of mobile equipment. In general, the EIR is used to check if mobile terminals have been stolen.

5. The future mobile systems

GSM has been a resounding success within Europe and the world. By offering a common standard for both air and network interfaces, operators have been able to provide digital cellular systems with extensive service capabilities. Despite the success of second generation systems, many limitations still exist. Developments are being made to define the future third generation systems which will cater for the mass-market mobile communications needs beyond the year 2000. Within Europe, this system is called universal mobile telecommunications system (UMTS), globally being referred to by the ITU as the future public land mobile telecommunications systems (FPLMTS).

The major aim of UMTS is to define an integrated system capable of providing a variety of mobile services (with a data rate of up to 2 Mbit/s) to encompass those currently offered by (dedicated) cordless, cellular and satellite mobile systems. UMTS will offer mass-market mobile communications to its users enabling access to services in the domestic, business and public (cellular) environments by using the same small 'pocketable' terminal.

5.1 Network requirements for UMTS

Mass-market fixed communications are currently provided by the PSTN GSM and first generation analogue systems offer cellular services, with cordless being widely used for tetherless access. The future telecommunications user will wish to access communications services from a common terminal. UMTS will be the step forward that enables this.

From the network evolution point of view, fixed networks are moving towards intelligent network (IN) based N-ISDN (narrowband ISDN) and B-ISDN (broad-

band ISDN) operation. Operators are adding intelligence into the network using INAP (intelligent network application part of SS No7) and SCPs (service control points). IN service creation and delivery techniques are also developing to enable network operators to offer a large variety of customised services.

The GSM mobile network is enabling digital mobile capabilities to be provided which are feature-rich in functionality. At present GSM mobile networks are providing services which are not widely available within fixed systems. As a stand-alone network GSM relies upon mobile specific network components (MSC/VLR and HLR). The same switching infrastructure cannot effectively be used for both fixed and mobile at present. The GSM mobility management and handover functions have been integrated into the switch (MSC). This has substantially increased some of the processing requirements to the switch, which in realistic terms has severely reduced the subscriber capability of GSM MSCs compared to equivalent fixed switches.

Due to the higher frequency of operation and large capacity requirement of UMTS, a dense cell architecture is required to provide adequate mass coverage; this will create large network infrastructure costs if the service, mobility management and handover features are developed around switch platforms in the same way as GSM. From the operators and equipment vendors perspective the stand-alone network approach is not feasible. The common aim for UMTS is harmonisation of the systems components via integration of mobile and fixed capabilities into a common architecture. This architecture consists of the core backbone switching network providing mobile call handling, with handover and mobility management being catered for by added intelligence — the intelligent network.

5.2 The IN solution for UMTS

The IN concept plays an important role in existing and future telecommunications systems. It is capable of providing simple, supplementary services that can be extended further to enable the implementation of advanced services and facilities in a flexible and efficient way. IN is seen as the solution for UMTS to enable network operators and equipment suppliers to produce integrated platforms which will cater for fixed narrowband and broadband communications and include mobile capabilities for UMTS. This minimises network infrastructure costs by reducing the number of network platforms within an operator's architecture while enabling common service provision for all end users, whether fixed or mobile. The IN approach will enable evolution of the GSM architecture to include the IN flexible service creation and delivery capabilities. A major aim of this approach is to enable UMTS to be added to net-

works via the addition of advanced intelligent capabilities to both the switching and the radio access components.

5.3 *IN capability set 1*

Within fixed networks, IN is being applied to provide a variety of functions, mainly related to the provision of simple telephony-based supplementary services. The IN capability set 1 (CS-1) standards use the service switching functions (SSF) and call control functions (CCF) to interact with the service control function (SCF) (which is supported by other functionality) to provide services. The CS-1 SCF is a single point of control and relies upon the SSF/CCF combination to invoke action. If added capabilities such as DTMF receivers, conference bridges or voice announcements (specialised resources) are needed, the SCF can switch in the specialised resource function (SRF) to provide these features. Although not fully achieved, CS-1 could be viewed as an architectural concept capable of providing additional capabilities to facilitate advanced service provision from a network-independent platform. To provide the advanced terminal mobility which is essential for UMTS, a variety of mobile-specific requirements must be satisfied.

5.4 *UMTS procedures*

As a mobile system, UMTS utilises many fixed and mobile procedures. Study of these procedures has identified a variety of requirements to the network operation.

Call handling

For mobile originating calls, the subscriber's service profile must be available within the local network (where the subscriber has roamed) for the visited network to verify the subscriber's requests for services and features. This will require inter-network communications to enable the profile transmission between the home and visited network.

Call handling for mobile terminating UMTS calls will use the separation of call and connection control. This is a concept introduced within broadband communications developments which is very applicable to mobile communications where the location and status of the called terminal may not be readily available when the call is originated. For UMTS terminated calls, the call processing should be halted at the earliest possible time and the locating procedure (to find the position of the called terminal) commenced. This will involve the switch-passing details of the called (UMTS terminal) number and the requested service to the SCF. The SCF will recognise that the dialled number is for a call destined to a UMTS number and will initiate locating by querying the called profile within the database (SDF). The profile holds details of the

current location, recorded status (busy, free) and details of the terminal and subscription capabilities. This enables the call to be failed, forwarded or routed at the earliest opportunity with the optimum use of the network resources. For a successful call, the SCF will communicate with the distant (visited network's) SCF and initiate paging of the called terminal. The outcome of the paging will be a network address for the called terminal which will be communicated back to the calling SCF. The calling SCF will instruct its local switch to route to the called terminal and call processing can continue.

Mobility management

Within IN CS-1, all procedures are call related, relying upon a state machine (the basic call state machine) within the switch to 'trigger' and invoke IN processing. UMTS mobility management procedures such as location registration, location updates and paging of called terminals can occur outside an active call, consequently alternative (non-call related) techniques to invoke the IN processing are required. The intelligent network must provide the control of these actions and enable the UMTS terminals to interact with the SCF entities directly. Obviously, the terminal to SCF communications must be secure to maintain network integrity. This direct application of terminal to SCF communication illustrates the far-reaching enhancements needed to IN for UMTS.

Handover

Handover is the terminal mobility procedure which is likely to place the most substantial impact upon the network aspects for UMTS. Mobile terminals send and receive traffic to and from a particular base station connected to the network when calls or signalling associations are in progress. Within contemporary systems handover is only used to satisfy speech and data quality requirements due to a changing radio path. Within UMTS a handover may also be initiated at the request of the user (e.g. for subscription related reasons) or the network operator (typically for network management purposes), irrespective of link quality.

Re-routings a call to a new base station is a significant task within the fixed network. Figure 3 shows a typical tree-like network topology supporting UMTS, in which the base station sub-system (BSS) comprises an arrangement of base stations and associated switching and control functionality. Often, handovers will take place between base stations of the same BSS, but they may also occur between base stations belonging to different BSSs, local exchanges (LE), or even transit exchanges (TX). During the process of handover, a common point has to be determined for bridging of the bearers between the old links and the new links, in order that the mobile terminal can make a seamless (unnoticeable) handover from the old to the new. As a consequence of the tree-like structure of the network, this

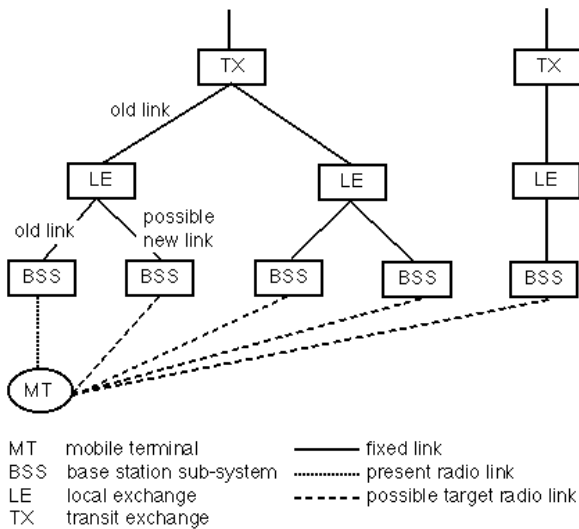


Fig 3 Handover possibilities within the network.

bridging point may be located at any level in the network. Thus handover has a large impact on the fixed network [3]. These impacts may be minimised if specific functionalities are split between the fixed network and control components. Use of IN control principles for handover within the core network will remove the processing burden of handover control integrated into the switches.

It is particularly important that handover is achieved rapidly (before a degrading radio channel is lost completely), and that the signalling load imposed on the network to provide handover is minimised.

5.5 Developments of IN for UMTS

To satisfy the needs of UMTS, and handover in particular, IN will have to evolve significantly from the current CS-1 position [4]. To provide UMTS, several major enhancements are required. The major developments are introduced below.

- Support of non-call related functions — certain mobility related functions (such as location management and handover) will require automatic invocation by the terminal when no call is in progress or without user intervention.
- Feature repeatability — the mobility features may need to be repeated several times during a call, e.g. a handover may be performed (at different levels of the network hierarchy) more than once.

- Concurrent and distributed execution of IN features — the UMTS terminal may need a number of simultaneous IN features (e.g. handover and location updating) to be performed. IN should be enhanced to support the execution of a variety of independent and concurrent IN-based services. The execution may occur at differing points within the network.
- Transfer of control — a UMTS terminal may move between the coverage areas of control entities during the execution of a service; this may require the service logic to be transferred mid-execution, or halted, transferred to another point of control and re-started. A typical example of this may be the handover procedure which will require interaction and co-operation between SCFs. If the handover of a call occurs between switches, the active BCSM (call state machine) will need to be passed from the old to the new switch. This should include conveyance of any active or armed trigger detection points.
- Advanced service interaction management capabilities — there are a number of areas where mobile features (such as handover) will have an impact on the provision of IN services. The contention between the handover procedure and call waiting supplementary service is one example of interaction that requires management. If the local exchange involved in the call must change due to handover it is essential that the IN service logic and switch-based detection point triggers catering for call waiting are also moved between switches during the handover. The interaction between SIBs responsible for handover and other IN services is easier to manage than an interaction between IN-based services and non-IN services.
- Intelligent terminal equipment — it must be possible to provide limited IN functionality in the mobile terminal, in order to support intelligent execution of handover and other UMTS services and procedures.

To achieve the full benefits of an IN approach to UMTS the evolving IN architecture requires distribution of intelligence within the core network, access network (base station system) and mobile terminal equipment.

5.6 The UMTS functional architecture

Figure 4 shows the IN-based functional architecture for UMTS [5]. The structure of the functional architecture for UMTS, including the allocation of functionality within it, is an essential step in defining the respective roles of IN service control and the supporting fixed network.

An important characteristic, which structures the architecture in line with IN concepts, is the separation of the UMTS service and mobility control, and data from the basic

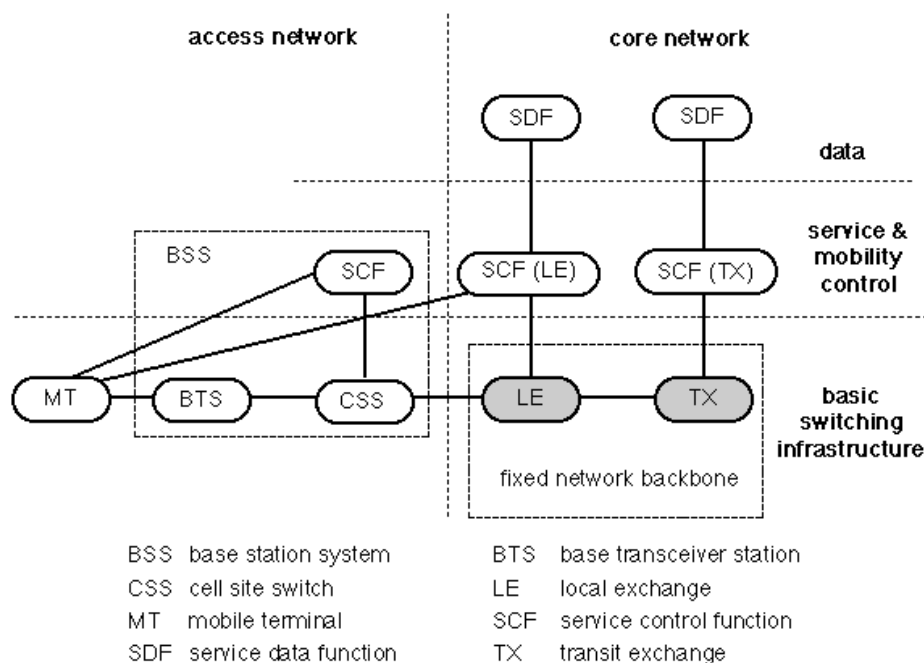


Fig 4 An IN-based UMTS functional architecture.

backbone switching infrastructure. The backbone network is represented by the LE and TX switching functional groups. These will meet the needs of both fixed and mobile communications services, but are not dedicated to UMTS alone.

A basic guideline, to ensure a generic backbone network, is that any UMTS-specific functionality should be removed from the LE/TX functional groups, as these represent basic switching capability. The traditional way of dealing with mobility procedures such as handover is to provide special mobility functionality in the switches. This, of course, ties the mobility functions to one particular network, thereby limiting scope for implementation, and results in the switches becoming UMTS-specific, rather than general purpose. For UMTS, the handover control functionality is therefore removed from the switching infrastructure and placed in the service control functions where it can be provided as an IN application.

The structure of the functional architecture is also influenced to a large extent by the requirements of the UMTS mobility procedures, and handover in particular. The main requirement of distributed control is reflected in the distributed nature of the network. SCFs at the BSS level as well as at the LE and TX level allow handover control functionality to be as close as possible to the bridging point, with the consequence that signalling links are as short as possible. The SCF has been distributed to cover the base station system, local exchange and transit exchange. This is essential in UMTS, where the time critical nature of some operations, such as handover, requires that functions are performed close to the point of invocation. The roles

performed by the individual SCFs may be very dependent upon their location, the core SCFs (relating to the switching) will include functionality very similar to contemporary IN CS-1 SCFs. The SCFc_{ss} and the CSS within the BSS will fulfil the role similar to the base station controller within GSM of providing intra BSS control and switching for handover purposes.

The SCFs found in the architecture will have the ability to process functions in a manner that will allow continuity of SIB chains across physical areas. This is illustrated in Fig 5 which shows the distribution of handover control functionality over different control functions within the network. The handover control can then be transferred between SCFs as the bridging point changes.

The functional architecture includes the mobile terminal (MT), which in UMTS will require a degree of IN functionality to enable support of advanced services such as handover. The MT represents one in a class of intelligent terminals which will carry limited service control functionality in order to support the advanced services of the future.

6. Conclusions

Mobile system developments to date have been reviewed, with emphasis placed upon the network components. The GSM architecture has been discussed with the future network requirements and developments for UMTS being outlined.

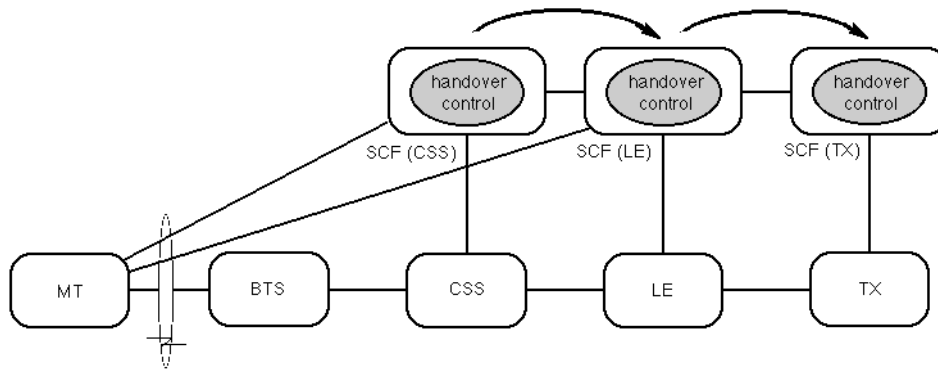


Fig 5 Transfer of handover control within the network.

The current stand-alone mobile and fixed architectures will be unable to cater for the major impacts that the future mass market mobile communications system, UMTS, will require. Significant enhancements are required to intelligent, fixed and mobile networks to provide the integrated communications network of the future. The intelligent network developments should include:

- non-call related processing,
- concurrent and distributed intelligence with transfer of control,
- the addition of more intelligent terminals to satisfy UMTS requirements.

Only if the UMTS requirements are included in the intelligent networks of the future will the significant burden of mobility management and handover be achievable. These can be provided within the same core switching architecture that can offer flexible service development and creation while minimising the major problem of services and features interaction.

With the inclusion of the network developments outlined, UMTS will be well placed to provide service to users (who are mobile and stationary) well into the twenty first century. This will have been achieved in the most efficient and cost-effective manner to both equipment vendors and network operators.

Acknowledgements

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Since moving to BT Laboratories in 1990 he has worked on mobile systems mainly contributing towards mobile and intelligent network developments for both GSM and UMTS. Recent work has included standardisation developments for UMTS within ETSI and participation in RACE projects on mobile networks.